

Cryptography Implementations

- Caesar Cipher

Plaintext: smartpole **Shift (K):** 3

Letter to Index Conversion:

Letter	s	m	a	r	t	p	o	l	e
Index (P)	18	12	0	17	19	15	14	11	4

Encryption Process using $C = (P + 3) \pmod{26}$:

Plain Index	18	12	0	17	19	15	14	11	4
After +3	21	15	3	20	22	18	17	14	7
Mod 26	21	15	3	20	22	18	17	14	7
Cipher	v	p	d	u	w	s	r	o	h

Cipher text: vpduwsroh

- Vigenère Cipher

Plaintext: smartpole **Key:** network

Encryption Process:

Plain	s	m	a	r	t	p	o	l	e
Key	n	e	t	w	o	r	k	n	e
P	18	12	0	17	19	15	14	11	4

Plai n	s	m	a	r	t	p	o	l	e
K	13	4	19	22	14	17	10	13	4
P+K	31	16	19	39	33	32	24	24	8
Mod 26	5	16	19	13	7	6	24	24	8
Ciph er	f	q	t	n	h	g	y	y	i

Cipher text: fqutnhgyi

• Rail Fence Cipher

Plaintext: smartpole Key (rails): 3

Zigzag Distribution:

Row / Pos	1	2	3	4	5	6	7	8	9
Row 1	s				t				e
Row 2		m		r		p		l	
Row 3			a				o		

Cipher text: stemrplao

Core Security Principles: The CIA Triad

1. Confidentiality

Definition: Confidentiality ensures that sensitive telemetry streams and system configurations are concealed from unauthorized individuals and only accessible to verified administrators.

What it does:

- Protects environmental logs (e.g., motion, gas, or sound sensor data) from network eavesdropping.
- Utilizes strong encryption protocols for data in transit and at rest.
- Enforces strict authentication mechanisms to verify node identity.
- Applies Role-Based Access Control (RBAC) to limit dashboard and administrative privileges.
- Mitigates the risk of unauthorized data exposure across the network infrastructure.

2. Integrity

Definition: Integrity guarantees that the transmitted data, routing tables, and device firmwares remain accurate, consistent, and completely unaltered by unauthorized entities.

What it does:

- Prevents man-in-the-middle modifications to sensor readings or control commands.
- Implements cryptographic hashing algorithms to verify payload authenticity.
- Detects physical tampering or logic errors at the microcontroller level.
- Ensures that analytical decisions are based on highly reliable and uncorrupted datasets.
- Protects system files and node configurations from unauthorized rewrites.

3. Availability

Definition: Availability ensures that the infrastructure network, localized nodes, and stored data logs remain reliably accessible to authorized systems whenever requested.

What it does:

- Maintains continuous uptime for critical sensing and logging nodes.
- Utilizes distributed load balancing to optimize processing across multiple servers.
- Defends the network architecture against Denial of Service (DoS) attacks.
- Employs redundancy and containerized backups for rapid disaster recovery.
- Ensures hardware constraints do not bottleneck systemic data flow.

Systems Risk Assessment

Definition: Risk Assessment is the structured methodology of cataloging network assets, identifying potential threats, evaluating inherent vulnerabilities, and implementing controls to safeguard the technical ecosystem.

Steps of Risk Assessment:

1. **Identify Assets:** Catalog all hardware (microcontrollers, sensors, gateways) and digital assets (telemetry logs, routing rules).
2. **Identify Threats:** Pinpoint external and internal dangers, such as unauthorized network intrusion, packet sniffing, or physical environmental damage.
3. **Identify Vulnerabilities:** Locate weaknesses, such as transmitting data over plaintext HTTP, open port configurations, or hardcoded API credentials.
4. **Analyze Risk:** Calculate the mathematical probability of a vulnerability being exploited against the potential impact on the infrastructure.
5. **Evaluate Risk:** Prioritize vulnerabilities, addressing critical points of failure before optimizing secondary system features.
6. **Mitigation (Control):** Deploy technical countermeasures, such as transitioning to HTTPS/MQTT-TLS, isolating node networks, and patching firmware.

What it does:

- Preemptively hardens the system against cyber-physical attacks.
- Minimizes the attack surface area of connected end-devices.
- Secures the flow of localized node data to centralized servers.
- Creates a resilient framework for sustained infrastructure deployment.

Practical Example:

- **Asset:** A deployed Smart Pole equipped with gas and sound sensor arrays.
- **Threat:** Malicious packet interception on the local network.
- **Vulnerability:** Unencrypted data streams between the sensor node and the central server.
- **Risk:** Exposure of localized environmental data and potential injection of false sensor triggers.